

Lab. 4
Part - 1

Student	Study hours	Pass (1) / Fail (0)
1	2	0
2	4	0
3	6	1
4	8	1
5	10	1

Step 1 → Possible split points for study hours.

Study hours → 2, 4, 6, 8, 10

$$\text{split points} \rightarrow \frac{(2+4)}{2} = 3$$

$$\frac{4+6}{2} = 5$$

$$\frac{6+8}{2} = 7$$

$$\frac{8+10}{2} = 9$$

So possible split points are 3, 5, 7 & 9

Step 2 - Gini impurity for resulting groups.

"Study Hours $\leq$ split", "Study Hours $>$ split"

$$Gini(p) = 1 - p_1^2 - p_0^2$$

Split at 3.

Left group (hours ≤ 3): {2}
Labels: {0}

Count: 1 instance

0 pass, 1 fail.

$$p_0 = 1/1 = 1$$

$$p_1 = 0/1 = 0$$

$$Gini-left = 1 - (0)^2 - (1)^2$$

$$= 0$$

Right group (hours > 3): {4, 6, 8, 10}

Labels: {0, 1, 1, 1}

Count: 4 instance

1 fail, 3 Pass

$$p_0 = 1/4 = 0.25$$

$$p_1 = 3/4 = 0.75$$

$$Gini-right = 1 - (0.25)^2 - (0.75)^2$$

$$= 1 - 0.0625 - 0.5625$$

$$= 0.375$$

$$Avg\ Gini = \frac{1}{5} \times 0 + \frac{4}{5} \times 0.375$$

$$= 0 + 0.3$$

$$Avg\ Gini = 0.3$$

split at 5:

Left group (hours ≤ 5): {2, 4}

Labels: {0, 0}

Count: 2 instance

0 pass, 0 fail

$$p_0 = 2/2 = 1$$

$$p_1 = 0/2 = 0$$

$$\text{gini-left} = 1 - (1)^2 - (0)^2 = 0$$

Right group (hours > 5): {6, 8, 10}

Labels: {1, 1, 1}

Count: 3 instance

3 pass, 0 fail

$$p_0 = 0/3 = 0$$

$$p_1 = 3/3 = 1$$

$$\text{gini-right} = 1 - (0)^2 - (1)^2 = 0$$

— Avg gini = ~~0~~

$$\text{Avg gini} = \frac{2}{5} \times 0 + \frac{3}{5} \times 0$$

$$= 0$$

Split at 7

Left-group (Hours ≤ 7): {2, 4, 6}

Labels: {0, 0, 1}

Count: 3 instances

1 pass, 2 fail

$$p_0 = 2/3 \approx 0.667$$

$$p_1 = 1/3 \approx 0.333$$

$$\text{Gini-left} = 1 - (0.667)^2 - (0.333)^2$$

$$= 1 - 0.444 - 0.111$$

$$= 0.445$$

Right-group (Hours > 7): {8, 10}

Labels: {1, 1}

Count: 2 instances

2 pass, 0 fail

$$p_0 = 0/2 = 0$$

$$p_1 = 2/2 = 1$$

$$\text{Gini-right} = 1 - (0)^2 - (1)^2 = 0$$

$$\text{Avg gini} = \frac{3}{5} \times 0.445 + \frac{2}{5} \times 0$$

$$= 0.267$$

Split at 9

Left group ($hour_i \leq 9$): {2, 4, 6, 8}

Labels: {0, 0, 1, 1}

Count: 4 instances

2 pass, 2 fail

$$p_0 = 2/4 = 0.5$$

$$p_1 = 2/4 = 0.5$$

$$Gini_{left} = 1 - (0.5)^2 - (0.5)^2$$

$$= 1 - 0.25 - 0.25$$

$$= 0.5$$

Right group ($hour_i > 9$): {10}

Labels: {1}

Count: 1 instance

1 pass, 0 fail

$$p_0 = 0/1 = 0$$

$$p_1 = 1/1 = 1$$

$$Gini_{right} = 1 - (0)^2 - (1)^2$$

$$= 0$$

$$Avg\ gini = \frac{4}{5} \times 0.5 + \frac{1}{5} \times 0$$

$$= 0.4$$

Step 3: Choose the split with lowest gini impurity

split point	Weighted Gini Impurity
3	0.300
5	0.000
7	0.267
9	0.400

Best split: study hours ≤ 5 (gini = 0.0)

Step 4: Draw the first-level decision tree

